The circle carries a self-homotopy that is not the reflexivity family

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 456-458 and its proof environment opens on line 459, the next line; the pairing is by adjacency and the retained proof range 459-468 is the whole environment.

Editorial note. The retained counter values (chapter 6, section 4, theorem offset 1) make the counted lemma print as Theorem 6.4.2, matching the original book numbering read from the label-number dump of the pinned snapshot (newlabel thm:S1-autohtpy = 6.4.2).

Editorial note. Three labels the proof cites lie outside the card and are declared in external_references as non-load-bearing navigation pointers: rmk:varies-along (book Remark 6.2.4, defined at line 203 of this chapter pinned file), sec:compute-paths (book section 2.11, whose file is not part of the pinned corpus so the pointer span is the label line of this chapter dump) and thm:loop-nontrivial (book Theorem 6.4.1, defined at line 444 of this chapter pinned file and counted by its own card of this lane).

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the circle has a self-homotopy other than the reflexivity family (author label thm:S1-autohtpy, book Theorem 6.4.2).

Allowed standard prerequisites (editorial).

- Induction along a loop. To define a dependent function over the circle it suffices to give its value at the base point and the dependent path required when the point varies along the generating loop.. It is how the self-homotopy is constructed.
- Transport in the loop family. Transporting a loop along a loop in the family x is a path from x to x computes as the inverse of one loop concatenated with the other and with the first again.. It turns the required dependent path into a concatenation identity.
- The generating loop is not trivial. The generating loop of the circle is not equal to the reflexivity path at the base point.. It distinguishes the constructed self-homotopy from the reflexivity family, and it is retained as a navigation pointer.

Source reference. Original remark 6.2.4: sources/hott-hits/pinned-original-source.tex, source lines 203–203. Book Remark 6.2.4, the rule for defining a dependent function on a higher inductive type along a path constructor, cited at source line 461 for the shape of the datum the construction needs. Navigation pointer only.

Source reference. Original section 2.11: sources/hott-hits/book-labelnumbers-first-edition.tex, source lines 1627–1627. Book section 2.11, where the transport in the loop family is computed, cited at source line 462. Navigation pointer only: the retained proof states the computed identity it uses. That section file is not part of the pinned corpus, so the pointer span is the label line of this chapter dump.

Source reference. Original theorem 6.4.1: sources/hott-hits/pinned-original-source.tex, source lines 444–444. Book Theorem 6.4.1, non-triviality of the generating loop, cited at the last line of the retained proof. Navigation pointer only: it is counted by its own card of this lane and nothing of the counted proof is delegated to it.

Lemma 6.4.2. *There exists an element of* $\prod_{(x:\mathbb{S}^1)}(x = x)$ *which is not equal to* $x \mapsto \mathsf{refl}_x$.

Proof. We define $f : \prod_{(x:\mathbb{S}^1)}(x = x)$ by $\mathbb{S}^1$-induction. When x is base, we let $f(\mathsf{base}) :\equiv \mathsf{loop}$. Now when x varies along loop (see original remark 6.2.4), we must show that $\mathsf{transport}^{x\mapsto x=x}(\mathsf{loop}, \mathsf{loop}) = \mathsf{loop}$. However, in original section 2.11 we observed that $\mathsf{transport}^{x\mapsto x=x}(p, q) = p^{-1} \cdot q \cdot p$, so what we have to show is that $\mathsf{loop}^{-1} \cdot \mathsf{loop} \cdot \mathsf{loop} = \mathsf{loop}$. But this is clear by canceling an inverse.

To show that $f \neq (x \mapsto \mathsf{refl}_x)$, it suffices to show that $f(\mathsf{base}) \neq \mathsf{refl}_{\mathsf{base}}$. But $f(\mathsf{base}) = \mathsf{loop}$, so this is just the previous lemma. □